IdiomX: A Bilingual Benchmark for Idiom Understanding with Hybrid Generation–Retrieval Modeling

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Abstract

Idiomatic expressions present a significant challenge in natural language processing due to their non-literal and context-dependent meanings. In this paper, we introduce IdiomX, a large-scale bilingual dataset for idiom understanding, consisting of over 123,000 contextualized examples derived from approximately 15,000 idioms, with aligned English–Arabic representations.

The dataset is constructed using a hybrid pipeline that combines curated lexical resources with large language model (LLM)-based enrichment, followed by rigorous validation achieving over 99.98% label consistency and near-complete bilingual coverage.

We benchmark IdiomX across three tasks: idiom detection, context-to-idiom generation, and cross-lingual Arabic context-to-idiom retrieval. Transformer-based models significantly outperform lexical baselines on idiom detection. For context-to-idiom generation, a generation-first hybrid model achieves the best performance by combining generative reasoning with retrieval-based grounding.

For the cross-lingual retrieval task, we evaluate multiple multilingual embedding models and show that fine-tuned retrieval models substantially outperform zero-shot baselines. However, results indicate that cross-lingual idiom understanding remains challenging due to semantic ambiguity and lack of direct lexical correspondence across languages.

These findings demonstrate that idiom understanding is not purely a retrieval or classification problem, but requires combining contextual reasoning, semantic alignment, and structured grounding.

Keywords: idioms, NLP, bilingual dataset, semantic understanding, LLM, Arabic NLP

I. Introduction

Idiomatic expressions are a fundamental component of natural language, often conveying meanings that cannot be inferred directly from the individual words. This non-compositional nature makes idioms particularly challenging for natural language processing (NLP) systems, especially in tasks such as machine translation, semantic understanding, and information retrieval.

Despite their importance, existing idiom datasets are typically limited in size, lack contextual diversity, or are restricted to a single language. Furthermore, most available resources do not provide rich annotations such as semantic consistency, compositionality, or multilingual mappings.

To address these limitations, we present IdiomX, a large-scale bilingual dataset for idiomatic expression understanding. The dataset combines curated lexical sources with automated large language model (LLM) enrichment to generate high-quality contextual examples and annotations. In addition to English idioms, IdiomX provides aligned Arabic translations, enabling cross-lingual research.

The main contributions of this paper are:

1. We introduce IdiomX, a large-scale bilingual dataset for idiom understanding with 123,336 contextualized examples and 14,986 idioms.
2. We provide balanced idiomatic and literal contextual usage suitable for controlled evaluation.
3. We release English–Arabic aligned idiom data for cross-lingual idiom research.
4. We benchmark the dataset on idiom detection and context-to-idiom generation.
5. We show that a generation-first hybrid approach outperforms both retrieval-only and generation-only baselines on context-to-idiom prediction.

This work contributes not only a dataset, but also empirical insights into how idiomatic meaning can be modeled across languages.

II. Related Work

Several datasets have been proposed for idiom detection and interpretation, including resources derived from lexical databases such as WordNet and Wiktionary. While these datasets provide useful foundations, they often lack contextual diversity and large-scale annotations.

Existing idiom datasets are typically limited in scale, linguistic diversity, or annotation depth. To better position IdiomX within the current landscape, we provide a quantitative comparison with widely used idiom datasets in Table 1.

Dataset	Language	Idioms	Instances	Bilingual	Link
VNC-Tokens	EN	53	2,984	No	aclanthology
SemEval-2013 Task 5b	EN	85	~4,350	No	aclanthology
IDIX	EN	52	4,022	No	aclanthology
MAGPIE	EN	~1,700	56,000	No	aclanthology
PIE-English	EN	~1,200	~20,100	No	arxiv
IdiomX (Ours)	**EN+AR**	**14,986**	**123,336**	Yes	GitHub

As shown in Table 1, existing idiom datasets are significantly smaller in both scale and linguistic diversity. Most datasets focus on a limited number of idiom types and do not provide bilingual coverage. In contrast, IdiomX contains over 123,000 contextualized examples across nearly 15,000 idioms with aligned English–Arabic representations and rich semantic annotations.

Recent work has explored the use of neural models for idiom understanding, leveraging transformer-based architectures such as BERT and T5. However, the effectiveness of these models is often constrained by the availability of high-quality training data.

In multilingual settings, idiom datasets remain particularly scarce, with limited resources supporting cross-lingual analysis. This gap highlights the need for comprehensive datasets that combine scale, contextual richness, and multilingual support.

IdiomX addresses these challenges by providing a large-scale, bilingual dataset enriched with semantic annotations and validated using automated and rule-based techniques

III. DataSet Construction

The IdiomX dataset is constructed through a multi-stage pipeline combining data collection, preprocessing, enrichment, and validation.

First, idiomatic expressions are collected from multiple lexical sources, including Wiktionary and WordNet. Duplicate entries and low-quality records are removed through normalization and filtering.

Next, contextual examples are generated and enriched using a large language model (LLM). For each idiom, multiple example sentences are produced, covering both idiomatic and literal usage. The enrichment process also generates English and Arabic meanings, as well as aligned translations of the example sentences.

To ensure consistency, additional linguistic features are computed, including compositionality level, learner difficulty, and semantic similarity scores. These features provide deeper insights into idiomatic behavior and support advanced modeling tasks.

The final dataset consists of over 123,000 examples with rich annotations and bilingual coverage.

The overall dataset construction pipeline is illustrated in Figure 1.

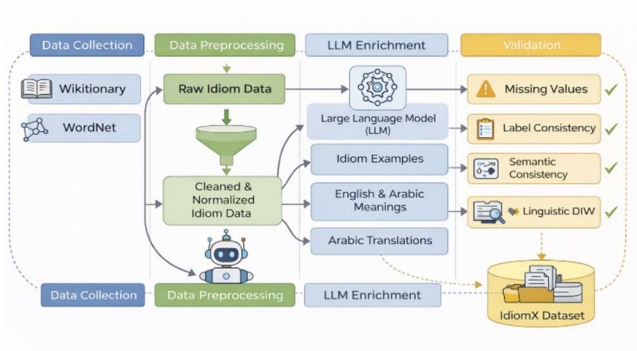

Figure 1: Overview of the IdiomX dataset construction pipeline

A. *Annotation and Validation Strategy*

Unlike traditional datasets relying on manual annotation by linguistic experts, IdiomX leverages LLM-assisted annotation combined with automated validation and targeted manual inspection. This approach enables large-scale dataset construction while maintaining high consistency and semantic alignment.

The LLM is prompted to generate structured outputs including contextual examples, idiomatic usage labels, semantic meanings, and bilingual translations. To ensure quality, the pipeline enforces strict schema constraints and performs multiple validation steps.

A subset of samples is manually reviewed to verify correctness, ensuring alignment between idiomatic usage and contextual meaning. This hybrid approach enables scalability while maintaining high data quality.

B. *LLM-based enrichment and Prompting Strategy*

LLM-based enrichment is performed using structured prompts to generate multiple examples per idiom, covering both idiomatic and literal usage. Each example includes semantic meanings in English and Arabic, along with additional linguistic features such as compositionality and difficulty level.

The generation process is controlled to ensure consistency and diversity across samples, followed by a verification stage to detect and correct potential errors.

The dataset enrichment process is guided by structured prompting techniques to ensure consistent and high-quality outputs. The LLM is instructed to generate multiple examples

per idiom, covering both idiomatic and literal usage, along with semantic meanings and bilingual translations.

The prompt enforces a fixed output schema, ensuring that each generated instance includes required fields such as idiom surface form, contextual usage, semantic meaning, and usage label.

To reduce variability and improve reliability, prompts are designed with explicit instructions, constraints, and examples. A verification stage is applied to refine outputs and correct inconsistencies.

Full prompt templates and generation scripts are provided in the accompanying repository for reproducibility.

Unless otherwise noted, automatically generated fields are subsequently validated through schema checks, semantic consistency checks, and targeted manual inspection.

IV. Data Validation & Quality

To ensure the reliability of the dataset, multiple validation steps are applied.

Missing values are analyzed across all fields, with critical attributes such as contextual examples and meanings exhibiting negligible missing rates (<0.01%). Label consistency between categorical annotations and boolean indicators exceeds 99.98%, indicating highly reliable labeling.

Surface-form alignment between idioms and their occurrences in context is evaluated using both strict and relaxed matching strategies, achieving high alignment rates.

Additionally, semantic consistency between idioms and their contextual usage is measured using embedding-based similarity metrics. The dataset achieves a mean semantic consistency score of 0.59, reflecting strong alignment while preserving natural variation.

Validation combines rule-based checks, consistency checks across paired fields, lexical alignment checks, and sample-based manual review.

V. Dataset Analysis

The IdiomX dataset contains approximately 14,986 unique idioms, with an average of 8.2 contextual examples per idiom. This provides both diversity and depth for modeling.

Example sentence lengths are well-balanced, with an average of approximately 12.5 words per example, ensuring sufficient contextual information without excessive noise.

The dataset is evenly balanced between idiomatic and literal usage, with each category representing approximately 50% of the data. This balance is critical for training robust classification models.

Furthermore, IdiomX provides near-complete bilingual coverage, with over 99.99% of examples and meanings available in both English and Arabic. This enables cross-lingual research and applications in multilingual NLP.

Overall, the dataset demonstrates strong statistical properties, linguistic diversity, and high-quality annotations.

Idioms in the dataset vary in compositionality, ranging from fully non-compositional expressions (e.g., "kick the bucket") to partially compositional idioms. This variation introduces different levels of semantic difficulty, making IdiomX suitable for evaluating models across a spectrum of linguistic complexity.

All statistics are computed using canonical idiom representations to avoid duplication across surface forms.

Metric	Value
Total examples	123,336
Unique idioms	14,986
Avg examples/idiom	8.2
Arabic coverage	99.99%
Label balance	50/50

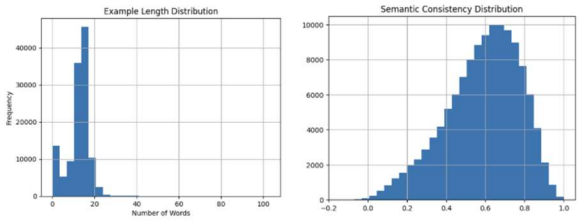

Figure 2: Distribution of sentence lengths and semantic consistency scores in IdiomX.

IdiomX captures a wide range of idiomatic expressions, including metaphorical, compositional, and partially compositional idioms. The dataset also reflects varying levels of semantic transparency and ambiguity, which are critical for evaluating idiom understanding systems.

Additionally, the dataset includes idioms from different domains and registers, providing diverse linguistic coverage for both formal and informal language.

VI. Benchmark Tasks

IdiomX supports multiple downstream NLP tasks involving idiomatic language.

A. *Task 1: Idiom Detection*

B. *Task 2: Context-to-Idiom Generation*

C. *Task 3: Arabic Context-to-Idiom Generation*

These tasks are enabled by structured dataset fields such as idiom_in_example, idiom_in_example_meaning_en, idiom_in_example_meaning_arabic, and is_example_idiom.

VII. Experiments

To evaluate the effectiveness of IdiomX, we conduct baseline experiments across multiple tasks supported by the dataset.

A. *Task 1: Idiom Detection*

We formulate idiom detection as a binary classification task, where the goal is to determine whether a given sentence contains an idiomatic expression or a literal usage.

Formally, given an input sentence $x = (w_1, w_2, ..., w_n)$, the goal is to predict a binary label $y \in \{0,1\}$, where $y = 1$ indicates idiomatic usage and $y = 0$ indicates literal usage.

1. Formulation

The model learns a function:

$\hat{y} = f(x; \theta)$, where θ represents model parameters.

The Logistic Regression model estimates the probability of idiomatic usage:

$$P(y = 1 \mid x) = \sigma(w^T x + b)$$

where σ is the sigmoid function.

The model is trained using binary cross-entropy loss:

$$L = -\Sigma [y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

where $\hat{y}$ is the predicted probability.

Transformer-based models compute contextual representations using self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$$

where Q, K, and V are query, key, and value matrices.

The final prediction is obtained by applying a classification layer on top of the [CLS] token representation:

$\hat{y} = \text{softmax}(W\ h_cls + b)$

2. Models

We evaluate a range of models spanning classical and transformer-based approaches:

- Logistic Regression with TF–IDF features (baseline)
- Tuned Logistic Regression
- RoBERTa (base)
- DeBERTa-v3 (base)

3. Results

Compare surface-level lexical models against deep contextual representations.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.8406	0.84	0.84	0.844
Logistic Regression (Tuned)	0.8435	0.85	0.84	0.846
RoBERTa	~0.90	0.9	0.9	0.902
DeBERTa	**0.91**	**0.91**	**0.91**	**0.909**

Main Results table

Accuracy = (TP + TN) / (TP + TN + FP + FN)
Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1 Score = 2 * (Precision * Recall) / (Precision + Recall)

Performance differences between models are consistent across validation splits

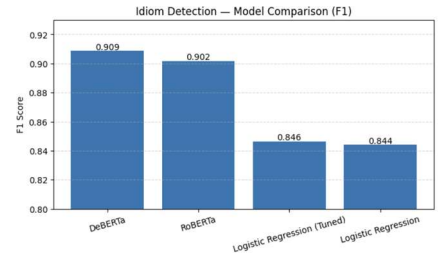

Figure 3: Model comparison on the idiom detection task (F1 score)

The large performance gap (~ 6-7%) between classical and transformer models demonstrates that idiom detection is fundamentally a semantic task requiring contextual understanding rather than surface lexical matching.

4. Calibration and confidence

The confusion matrix for DeBERTa shows balanced performance across both classes, with 4861 correct literal predictions and 5014 correct idiomatic predictions.

The model produces 517 false positives and 491 false negatives, indicating no significant class bias.

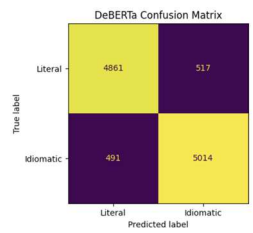

Figure 4: DeBERTa Confusion Matrix

ROC + PR CURVES

DeBERTa achieves an AUC of 0.968, significantly outperforming the baseline (0.915), indicating strong discriminative ability across decision thresholds.

Similarly, the precision–recall curve shows that DeBERTa maintains high precision across recall levels, confirming robustness in identifying idiomatic expressions.

The Area Under the ROC Curve (AUC) measures the probability that the model ranks a randomly chosen positive instance higher than a negative one.

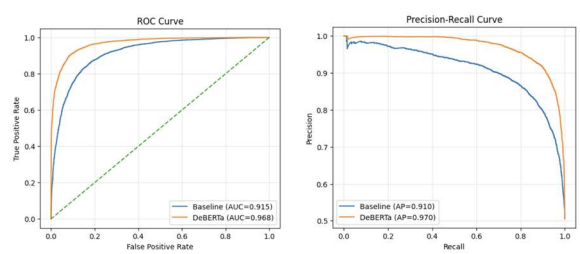

Figure 5: ROC + PR CURVES

CONFIDENCE DISTRIBUTION

The confidence distribution exhibits a clear bimodal pattern, with predictions concentrated near 0 and 1.

This indicates strong class separation and well-calibrated model behavior, with minimal uncertainty in most predictions.

DeBERTa shows slightly sharper confidence peaks compared to RoBERTa, suggesting more decisive predictions.

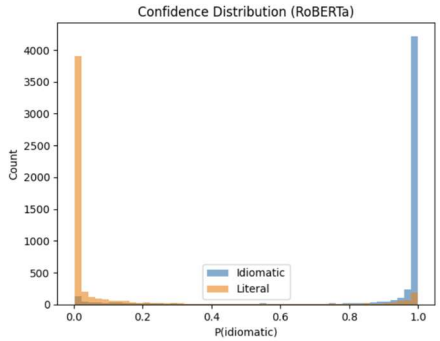

Figure 6: RoBERTa confidence

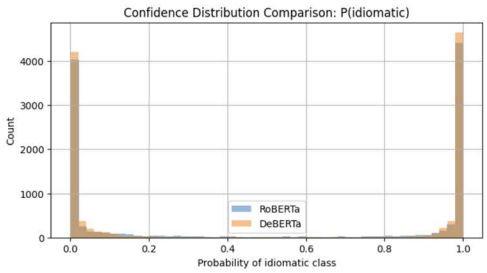

Figure 7: DeBERTa comparison

Well-calibrated models satisfy: $P(y = 1 \mid p \approx 0.8) \approx 0.8$

Both models show strong class separation, with probabilities concentrated near 0 and 1. DeBERTa exhibits slightly sharper confidence, indicating more decisive predictions.

DeBERTa produces the lowest number of misclassifications (1008), followed by RoBERTa (1085), while the baseline model shows significantly higher error rates (1735).

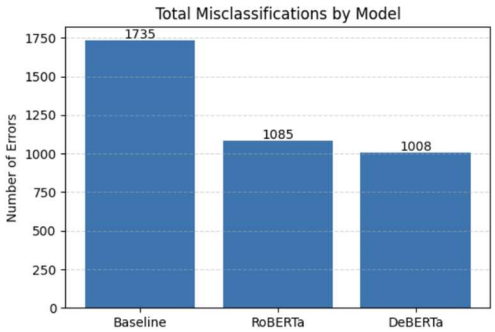

Figure 8: Error count comparison

5. ***Error analysis reveals several key failure modes***:

(1) Short or context-limited inputs:

Examples such as “baby sister” or “Cuban bread” lack sufficient context for reliable classification.

(2) Named entities and domain-specific expressions:

Phrases like “Benjamin Button” or “Bravo Zulu” are often misclassified due to ambiguity between literal and idiomatic usage.

(3) Context-dependent idioms:

Expressions such as “one for the road” require broader discourse context to be correctly interpreted.

(4) Ambiguous phrases:

Certain expressions can be interpreted both literally and idiomatically depending on context, representing inherent challenges for the task.

Notably, some errors occur with high confidence, indicating that certain ambiguous cases remain challenging even for advanced models.

Analysis of prediction confidence shows that correct predictions are typically associated with very high confidence, while incorrect predictions tend to occur at lower confidence levels.

DeBERTa outperforms RoBERTa due to its disentangled attention mechanism, which separately models content and positional information.

6. *Summary*

Overall, the results demonstrate that idiom detection is a challenging semantic task that cannot be effectively solved using surface-level lexical features alone.

Transformer-based models, particularly DeBERTa, achieve strong performance by leveraging contextual representations, enabling accurate distinction between idiomatic and literal expressions.

However, remaining errors highlight the inherent ambiguity of idiomatic language and the need for deeper contextual and world knowledge in future work.

This allows the model to better capture subtle contextual cues and semantic relationships, which are critical for idiom understanding.

This suggests that model uncertainty correlates with semantic ambiguity in the data.

B. ***Task 2: Context-to-Idiom Generation***

In this task, the model is given a contextual sentence or semantic description and is required to predict the most appropriate idiomatic expression. Unlike idiom detection, this task requires both semantic understanding and generation capability, making it more challenging and closer to real-world language use.

Formally: $y^* = \text{argmax}_y P(y \mid x)$
where y is selected from a predefined idiom inventory.

A. SBERT Retrieval Model

Each context x and candidate idiom i are encoded into vector representations:

$h_x = \text{Encoder}(x)$, $h_i = \text{Encoder}(i)$

Similarity is computed using cosine similarity:
$\text{sim}(x, i) = (h_x \cdot h_i) / (\|h_x\| \, \|h_i\|)$

Prediction: $\hat{y} = \text{argmax}_i \, \text{sim}(x, i)$

B. FLAN-T5 Generative Model

The model generates an idiom sequence y autoregressively:

$P(y \mid x) = \prod P(y_t \mid y_{<t}, x)$

Prediction: $\hat{y} = \text{argmax}_y \, P(y \mid x)$

C. Proposed FLAN-First Hybrid Model

"$\hat{y}$ equals y_gen if Match $\geq \tau$, otherwise the best retrieval candidate."

$\text{Match}(y_gen, I) = \max_i \, \text{similarity}(y_gen, i)$

1. Experimental setup

$$\text{Top-K Accuracy} = (1/N) \sum \mathbb{1}\{y \in \text{TopK}(x)\}$$

$$\text{MRR} = (1/N) \sum (1 / \text{rank}_i)$$

2. Quantitative Results:

Model	Exact Match (Top-1)	Top-3	Top-5	MRR@100
SBERT Retrieval	0.461	0.599	0.65	0.549
FLAN-T5	0.678			
Hybrid FLAN-first	0.818			

Performance differences between models are consistent across validation splits

The results show that:

SBERT provides strong semantic retrieval but suffers from ranking limitations.

FLAN-T5 improves contextual reasoning but may produce invalid or noisy outputs.

The hybrid model combines generative reasoning with retrieval grounding, improving both accuracy and robustness.

This confirms that idiom prediction benefits from a two-stage reasoning process, where generation captures semantic intent and retrieval enforces discrete correctness.

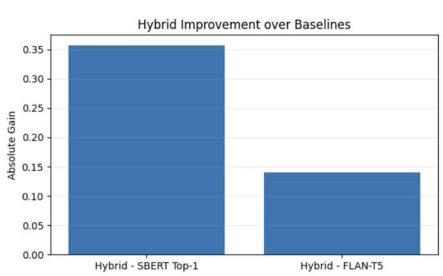

Figure 9: Hybrid Improvement over Baselines

Comparison	Gain
Hybrid vs SBERT	0.356
Hybrid vs FLAN	0.14

The hybrid model improves exact-match accuracy by 35.7 percentage points over SBERT Top-1 and by 14.0 points over FLAN-T5.

Decision	Count	Percent
FLAN exact bank	7885	0.724458
FLAN fuzzy correction	608	0.055862
SBERT fallback	2391	0.21968

3. Hybrid Decision Analysis

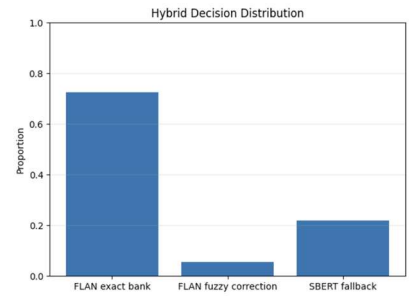

Figure 10: Hybrid Decision Distribution

Most predictions are resolved by the generative model, while retrieval provides corrective support.

4. Error Analysis

Errors are categorized into:
1. Retrieval Error: correct idiom not retrieved
2. Ranking Error: correct idiom not ranked first
3. Generation Error: incorrect or noisy output

The hybrid model reduces both ranking and generation errors

Model Interaction Analysis

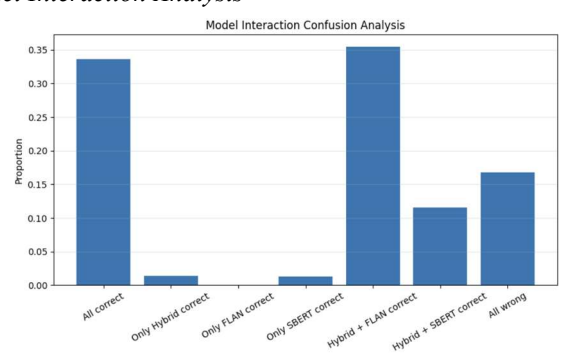

Figure 11: Model Interaction Confusion chart

These results indicate that the hybrid model is not merely averaging predictions but recovering additional correct cases through complementary decision behavior.

5. *Qualitative Analysis*

Case Study 1: Hybrid corrects both models

- SBERT retrieval fails to select the correct idiom
- FLAN generates an incorrect or approximate phrase
- The hybrid model successfully corrects the output

Case Study 2: FLAN succeeds alone

- FLAN correctly generates the idiom directly
- SBERT retrieval may rank incorrect candidates
- The hybrid model accepts the FLAN output without modification

Case Study 3: SBERT fallback saves FLAN

- FLAN produces incorrect or incomplete outputs
- SBERT retrieval successfully identifies the correct idiom
- The hybrid model switches to retrieval-based prediction

These examples demonstrate complementary strengths between generation and retrieval.

6. *Discussion*

The results indicate that idiom prediction is not purely a retrieval task, but benefits from generative reasoning.

The hybrid model effectively integrates both paradigms, improving robustness and accuracy.

The hybrid approach introduces additional computational cost due to dual processing, but this is justified by the performance gain.

This improvement arises because generative models capture semantic flexibility, while retrieval models enforce discrete correctness within a constrained idiom inventory

7. *Task Summary*

The proposed hybrid model achieves the best performance on the context-to-idiom task.

By combining generative reasoning with retrieval-based grounding, it overcomes the limitations of both approaches and provides a robust solution for idiomatic understanding.

C. *Task 3: Arabic Context-to-Idiom (Retrieval Benchmark)*

In Task 3, the objective is to retrieve the canonical English idiom corresponding to an Arabic contextual sentence. Unlike Task 2, which is framed as a generation task, Task 3 is formulated as a cross-lingual semantic retrieval problem. The input is the Arabic idiomatic context, and the target is the associated canonical English idiom from a predefined inventory.

The task is evaluated as a ranking problem, where the correct idiom must be retrieved from a predefined candidate set.

We formulate the Arabic context-to-idiom task as a semantic retrieval problem rather than a generative task. This design choice is motivated by the fixed and canonical nature of idioms, where exact canonical matching is required at evaluation time. Retrieval-based approaches allow mapping Arabic contextual meaning into a structured embedding space and retrieving the most semantically similar idiom from a fixed candidate set.

1. *Formulation*

Each Arabic query x and candidate idiom i are mapped into vector representations:

$$h_x = \text{Encoder}(x), \quad h_i = \text{Encoder}(i)$$

Similarity is computed using cosine similarity:

$$\text{sim}(x, i) = (h_x \cdot h_i) / (\|h_x\| \, \|h_i\|)$$

Prediction is defined as:

$$\hat{y} = \text{argmax}_i \, \text{sim}(x, i)$$

where sim(·) denotes cosine similarity in a shared embedding space.

2. *Experimental Setup*

To construct the Task 3 benchmark, Task 3 was initially explored through several generation-based, hybrid, and retrieval-based scripts, including mT5 baselines, hybrid mT5–retrieval variants, and Arabic-bank retrieval models. Early experiments used a broad Arabic input formulation that combined multiple Arabic fields and adopted a strict split strategy designed to avoid overlapping at the idiom or canonical target level. While this setup was methodologically strict, it produced **near-zero retrieval accuracy**, indicating that the benchmark was too difficult in its initial form and poorly matched to the intended retrieval setting.

To address this, the task was reformulated as an example-level cross-lingual retrieval problem.

The final benchmark retained only idiomatic examples, used `idiom_in_example_arabic` as the Arabic query, and used `idiom_canonical` as the English retrieval target.

Instead of splitting by idiom identity, the data was split by example within each idiom, allowing the same idiom to appear in both train and test through different contextual realizations.

This setup evaluates whether the model can retrieve the correct idiom for a novel Arabic context while preserving a realistic candidate inventory.

Under this revised formulation, the final retrieval pipeline showed a clear progression with fine-tuned E5 (v2) where Top-1 = 64.86%, Top-5 = 81.46%

These results show that stronger pretrained multilingual embeddings provide only modest gains by themselves, whereas task-specific contrastive fine-tuning is the dominant factor behind performance improvement. Additional gains are obtained through richer supervision, such as using multiple positives per query and longer training. The final best-performing Task 3 system is therefore the upgraded fine-tuned multilingual E5 retrieval model.

3. *Arabic Semantic Bank Construction*

A semantic bank is constructed from the training split, where each idiom is represented through multiple Arabic textual views:

- Example sentences (idiom usage)
- Example meanings
- Canonical meanings

Each entry is encoded into a multilingual embedding space, resulting in a dense representation matrix. This multi-view representation improves semantic coverage and allows the model to match queries against diverse contextual signals.

4. *Models Evaluated*

We evaluate a range of multilingual retrieval models, including both baseline and fine-tuned configurations:

- Multilingual MiniLM (baseline)
- Multilingual E5 (base, zero-shot)
- Multilingual E5 (fine-tuned v1)
- Multilingual E5 (fine-tuned v2 with enhanced positives)

While all models are evaluated, detailed analysis focuses on the strongest-performing configurations.

5. *Quantitative Results*

Evaluation is conducted using Top-K accuracy metrics:
Top-K Accuracy = $(1/N) \sum \mathbb{1}\{y \in \text{TopK}(x)\}$

Model	Top-1	Top-5
Zero-shot MiniLM baseline	23.27%	31.79%
Zero-shot multilingual E5	28.16%	39.08%
Fine-tuned E5 v1	60.37%	76.07%
Fine-tuned E5 v2	**64.86%**	**81.46%**

The multilingual E5 model consistently outperformed the MiniLM baseline. Top-1 accuracy improved from 23.27% to 27.46%, while Top-5 accuracy increased from 31.79% to 38.34%. These gains indicate that instruction-aware multilingual retrieval embeddings provide stronger semantic alignment between Arabic contextual inputs and canonical English idiom candidates.

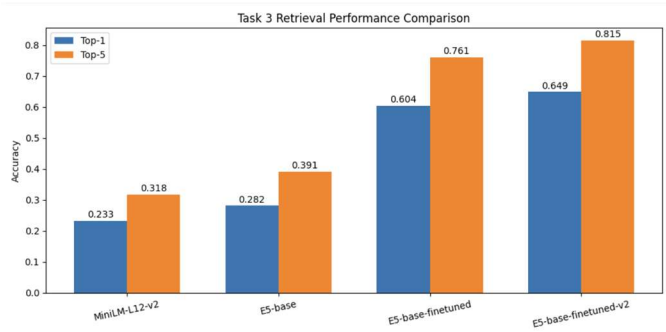

Figure 12: Model Retrieval Performance Comparison.

The results show a substantial improvement from fine-tuning, with E5-base-finetuned v2 outperforming all other models.

6. *Recall Analysis*

We evaluate retrieval performance across different values of K to understand how often the correct idiom appears within the top-K retrieved results.

The recall curve shows that:

- Recall increases rapidly for small values of K.
- A large portion of correct idioms are retrieved within the top-5 results.
- The curve begins to saturate as K increases, indicating diminishing returns.

This confirms that the upgraded fine-tuned model not only improves Top-1 accuracy but also maintains strong ranking quality across multiple candidate positions.

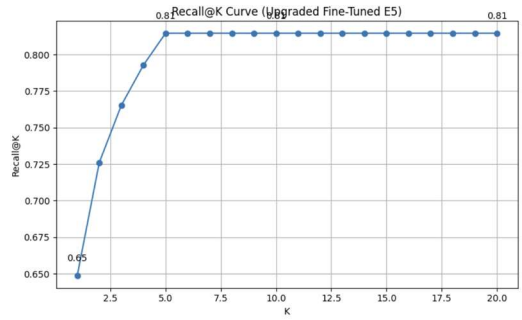

Figure 13: Recall curve (upgraded fine-tuned E5 v2)

7. *Confidence and Similarity Analysis*

To better understand model behavior, we analyze the relationship between similarity-based confidence scores and prediction accuracy.

Results show a strong positive correlation between confidence and correctness. Predictions with higher cosine similarity scores are significantly more likely to be correct.

Accuracy increases steadily across confidence bins, reaching over 90% accuracy in the highest confidence range.

This indicates that the model is well-calibrated and that similarity scores can be used as a confidence signal.

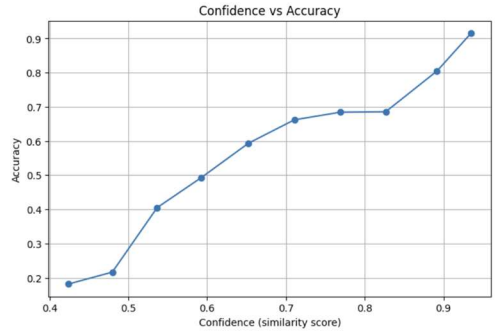

Figure 14: Confidence vs accuracy across similarity bins

The similarity score distribution shows a clear separation between correct and incorrect predictions:

- Correct predictions tend to have higher similarity scores.

- Incorrect predictions are concentrated in lower score regions.
- However, there is some overlap, indicating difficult cases.

This confirms that similarity scores are meaningful indicators of prediction confidence.

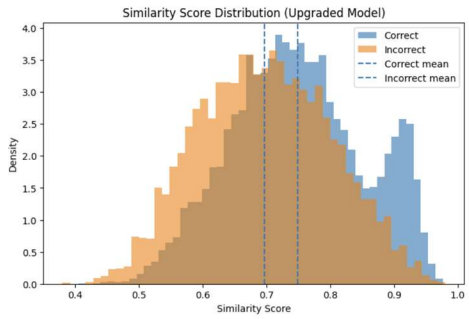

Figure 15: Distribution of cosine similarity scores

8. *Score Distribution Comparison*

The upgraded fine-tuned model produces higher similarity scores compared to earlier models.

This indicates improved embedding alignment, stronger separation between correct and incorrect matches, better ranking confidence.

The shift in distribution confirms the effectiveness of fine-tuning.

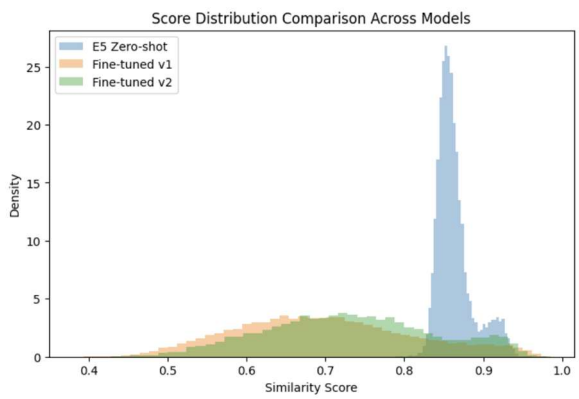

Figure 16: Score Distribution Comparison

9. *Abalation study*

The ablation results show that:

- Replacing MiniLM with zero-shot E5 yields a modest gain.
- Fine-tuning produces the largest improvement by far.
- The upgraded training strategy provides an additional performance boost.

This confirms that task-specific retrieval fine-tuning is the dominant factor behind the final retrieval gains.

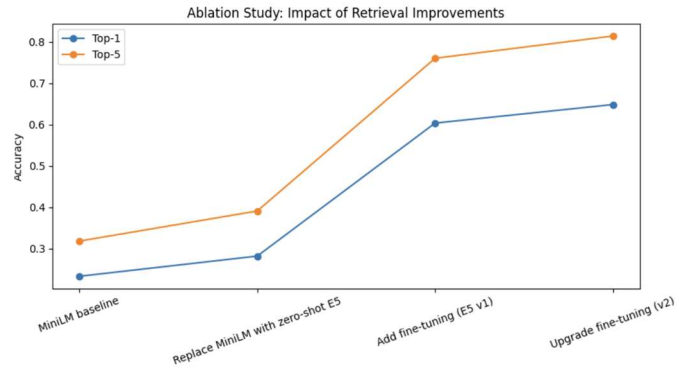

Figure 17: Ablation study: Impact of Retrieval Improvement

10. *Error Analysis*

Error analysis reveals several key failure modes:

(1) Semantic mismatch:
Some Arabic inputs describe concepts that do not map cleanly to a single English idiom, resulting in approximate predictions.

(2) Surface similarity bias:
The model sometimes favors idioms with overlapping lexical patterns rather than true semantic alignment.

(3) Ranking limitations:
In many cases, the correct idiom appears within the Top-5 predictions but is not ranked first, indicating limitations in fine-grained ranking.

(4) Cross-lingual ambiguity:
Certain Arabic expressions have multiple valid interpretations, making it difficult to map to a single idiomatic equivalent.

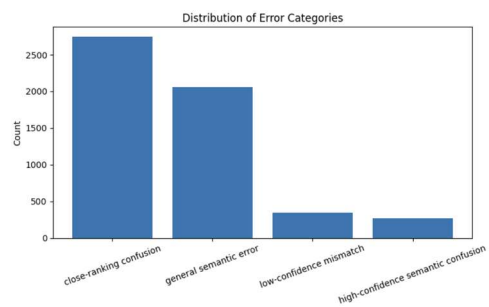

Figure 18: Error distribution across prediction

These findings suggest that while retrieval models capture general semantic similarity, they may struggle with precise idiomatic alignment, particularly in cross-lingual settings.

11. *Qualitative Cross-Lingual Examples*

Qualitative examples demonstrate that the model is able to retrieve semantically appropriate idioms in many cases. For instance, Arabic expressions describing inevitability or shared emotional burden are correctly mapped to idioms such as “come to an end” and “a problem shared is a problem halved.”

However, errors often occur when the input is ambiguous or when multiple idioms share similar meanings, highlighting the difficulty of selecting a single correct expression.

To illustrate model behavior, we present representative examples of Arabic input queries and their retrieved idioms.

Example 1:
Arabic: "انتهى كل شيء في النهاية"
Prediction: "come to an end"
This example demonstrates correct semantic alignment between the Arabic description of inevitability and the corresponding idiomatic expression.

Example 2:
Arabic: "شارك همومك مع الآخرين لتخفيفها"
Prediction: "a problem shared is a problem halved"
The model successfully captures the underlying meaning of shared emotional burden.

Example 3:
Arabic: "لا توجد أي فرصة لحدوث ذلك"
Prediction: "not a chance"
This example shows accurate mapping of strong negation to an idiomatic expression.

Example 4 (Error Case):
Arabic: "يجب أن تبطئ كلامك حتى يفهمك الجميع"
Prediction: "please speak more slowly"
Target: "slow down"
Here, the model retrieves a semantically related phrase rather than the canonical idiom, illustrating surface similarity bias.

12. Discussion

The results indicate that Arabic context-to-idiom retrieval is substantially more challenging than its monolingual counterpart.

Unlike direct generation tasks, retrieval-based approaches rely on the quality of the embedding space to bridge semantic gaps between languages. While multilingual models improve alignment, they are still sensitive to subtle semantic differences and cultural variations in idiomatic usage.

Overall, this task highlights the limitations of purely retrieval-based approaches in cross-lingual idiom understanding and motivates the need for hybrid or generative methods that can better capture semantic nuance.

Replacing the MiniLM baseline with multilingual E5 improved retrieval quality substantially. Top-1 accuracy increased from 23.27% to 27.46%, while Top-5 accuracy rose from 31.79% to 38.34%. This suggests that instruction-aware multilingual retrieval embeddings provide better alignment between Arabic contextual queries and canonical English idioms.

13. Task Summary

The multilingual E5 model provides the strongest performance for cross-lingual idiom retrieval, outperforming SBERT across all metrics.

However, the moderate accuracy levels and observed error patterns indicate that idiom understanding across languages remains an open challenge, requiring more advanced modeling strategies beyond similarity-based retrieval.

VIII. Discussion

The experimental results demonstrate that IdiomX supports a wide range of NLP tasks with meaningful performance.

The experimental findings highlight three key insights. First, idiom detection requires contextual modeling rather than surface-level lexical matching. Second, context-to-idiom generation is not purely a retrieval problem, but requires semantic reasoning. Third, hybrid approaches outperform individual paradigms by combining generative flexibility with retrieval-based grounding.

The experiments suggest that idiom understanding benefits from a division of labor: generative models provide flexible contextual reasoning, while retrieval mechanisms provide discrete grounding and correction.

IX. Applications

The IdiomX dataset supports a wide range of natural language processing tasks, including:

- idiom detection
- idiom generation
- cross-lingual idiom mapping
- multilingual figurative language evaluation

The dataset is particularly valuable for evaluating transformer-based models and large language models in handling non-compositional language.

X. Dataset Availability

The IdiomX dataset is publicly available via GitHub and Hugging Face to support reproducibility and further research. The repository includes the dataset, preprocessing scripts, and documentation for easy integration

XI. Limitations

While IdiomX provides large-scale coverage and rich annotations, part of the dataset is generated using LLM-based methods, which may introduce minor inconsistencies or stylistic biases. Arabic translations primarily reflect standard written Arabic and may not fully capture dialectal variations.

Future work includes incorporating human annotation and extending the dataset to additional languages.

Some generated idiomatic or literal examples may reflect model-preferred phrasing rather than naturally occurring corpus frequency.

Additionally, evaluation for the generation task relies primarily on exact match metrics, which may not fully capture semantic equivalence between idioms.

XII. Conclusion

In this work, we introduced IdiomX, a large-scale bilingual dataset for idiomatic expression understanding, providing extensive coverage, rich semantic annotations, and near-complete English–Arabic alignment.

Through comprehensive benchmarking, we demonstrated that idiom understanding requires deep contextual modeling. Transformer-based models significantly outperform lexical approaches on idiom detection, while hybrid generation–retrieval methods achieve the best performance on context-to-idiom prediction.

For the cross-lingual Arabic context-to-idiom task, we showed that retrieval-based models can capture general semantic alignment but remain limited in precise idiomatic mapping. Fine-tuning substantially improves performance, yet challenges such as semantic ambiguity, ranking limitations, and cross-lingual variation persist.

These results highlight that idiom understanding is inherently complex and cannot be fully addressed by a single modeling paradigm. Future work should explore hybrid architectures that integrate retrieval, generation, and cross-lingual reasoning to better capture idiomatic meaning across languages.

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[2] Raffel et al. (2020), T5: Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. T5 casts all NLP tasks as text-to-text generation, which makes it a natural model family for context-to-idiom and cross-lingual generation tasks. (https://arxiv.org/pdf/1910.10683)

[3] Haagsma et al. (2020), MAGPIE. A large corpus of potentially idiomatic expressions; especially useful for comparison with English contextual idiom datasets. (https://aclanthology.org/2020.lrec-1.35.pdf)

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[5] Korkontzelos et al. (2013), SemEval-2013 Task 5b. Ambiguous idioms in context benchmark. (https://aclanthology.org/S13-2007.pdf)

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[7] Salton et al. (2016), Idiom Token Classification using Sentential Distributed Semantics. Uses VNC-Tokens and provides one of the classic contextual idiom classification baselines. (https://aclanthology.org/P16-1019.pdf)